



SRI BALAJI COLLEGE OF ENGINEERING & TECHNOLOGY

Benad Road, Near Dadi Ka Phatak, Jaipur – 302013 (Rajasthan)

Approved by AICTE, New Delhi | Affiliated to Rajasthan Technical University (RTU), Kota

DEPARTMENT OF APPLIED SCIENCES & FIRST YEAR ENGINEERING

COMPREHENSIVE PROJECT REPORT & TECHNICAL SPECIFICATION

AttendIQ

Next-Generation Smart College Attendance Management Platform
with Dynamic QR Tokens, Campus Geofencing, Biometric Liveness
Verification, and Multi-Tier Institutional Analytics

Academic Session: 2026–2027

Target Degree: B.Tech (All Branches)

Software Release: v2.4-Production

Architecture: Progressive Web App (PWA)

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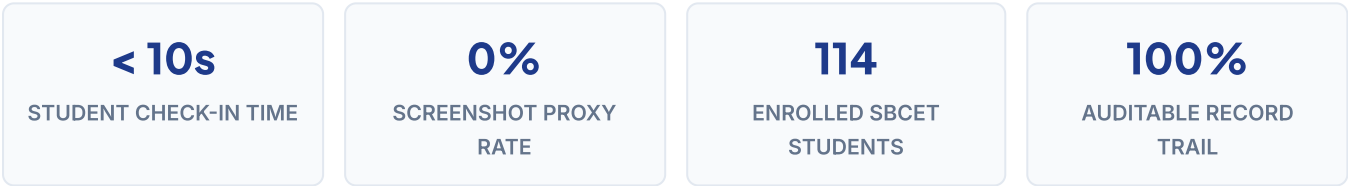
Academic Coordinators, First Year Engg.

EXECUTIVE SUMMARY

Abstract & Key Metrics

Traditional classroom attendance in Indian higher education institutions relies heavily on paper register sheets or static paper roll-calls, consuming 10–15 minutes of every 50-minute instructional period (~25% of active learning time). Even rudimentary digital systems that display static QR codes fail catastrophically due to "proxy attendance" enabled by instant screenshot sharing over messaging apps (WhatsApp, Telegram).

AttendIQ is a production-hardened, multi-tier Progressive Web App (PWA) engineered specifically for **Sri Balaji College of Engineering & Technology (SBCET), Jaipur**. AttendIQ eliminates proxy attendance through a synchronized four-stage verification pipeline: (1) **Short-lived dynamic QR tokens** rotating every 25 seconds with cryptographic entropy; (2) **GPS Haversine geofencing** enforcing physical student presence within a configurable campus radius (50m–500m) with fallback Campus Wi-Fi Beacon verification; (3) **In-browser camera facial recognition** matched against enrollment records with randomized interactive liveness challenges (blink, smile, head-turn) to defeat photographic presentation attacks; and (4) **Atomic composite key uniqueness locks** prohibiting duplicate check-ins.



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CHAPTER 1

Project Overview & Objectives

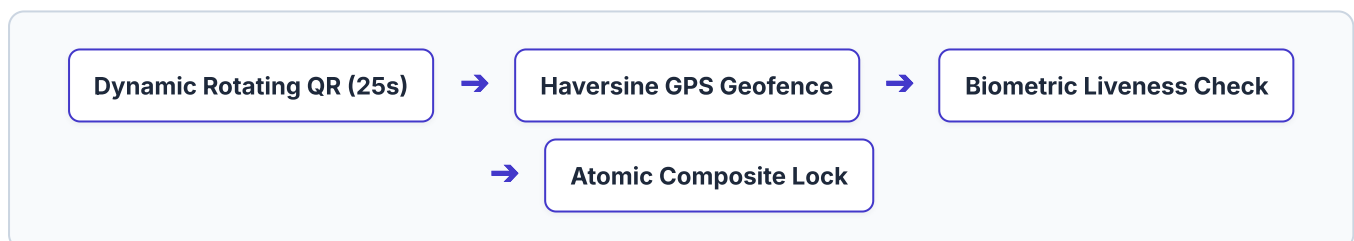
1.1 Institutional Background

Sri Balaji College of Engineering & Technology (SBCET), Jaipur, established in 2000, conducts undergraduate engineering degree programs affiliated with Rajasthan Technical University (RTU), Kota. RTU statutory guidelines mandate that students maintain a minimum of **75% attendance** in each theory and laboratory subject to be eligible to sit for end-semester university examinations.

1.2 Existing Bottlenecks & Vulnerabilities

- **Instructional Time Depletion:** Faculty spend between 10 to 15 minutes of every 50-minute lecture verbally calling roll numbers, reducing semester curriculum coverage by up to 25%.
- **Buddy Punching & Proxy Signatures:** In large lecture halls (65+ students in Section A), students routinely sign paper sheets on behalf of absent peers or whisper roll calls.
- **Digital Proxy via Screenshot Sharing:** First-generation QR systems display static QR codes. Absent students sitting in hostels or off-campus receive instant photos via WhatsApp and mark attendance remotely.
- **Administrative Delays & Manual Register Compilation:** Calculating monthly attendance percentages across 6 branches and 114 students requires days of manual clerical calculation.

1.3 Core Engineering Objectives



- **Sub-10-Second Frictionless Check-In:** Enable students to self-verify attendance seamlessly on mobile or desktop without instructor interruption.
- **Zero-Tolerance Anti-Proxy Enforcement:** Cryptographic dynamic token rotation defeats screenshot forwarding; GPS Haversine calculations defeat off-campus submissions; biometric liveness defeats photo presentation attacks.
- **Automated Institutional Timetable Integration:** Automatically link attendance sessions with scheduled lecture halls, sections, faculties, and time slots.
- **Real-Time 75% Statutory Shortage Alerts:** Provide automated graphical alerts to students and faculty whenever cumulative attendance drops below the 75% RTU threshold.
- **Official PDF Roll-Call & Register Generation:** Instant, client-side generation of printable official attendance sheets with faculty and HOD signature blocks.

CHAPTER 2

Complete Technology Stack & Architecture

AttendIQ is designed using modern single-page application (SPA) principles with full offline-capable Progressive Web App (PWA) architecture. The application is completely decoupled into presentation, service, domain store, and security layers.

2.1 Comprehensive Technology Stack

LAYER	TECHNOLOGY / PACKAGE	ARCHITECTURAL RATIONALE
Core Frontend	React 18.3.1	Component-driven architecture, concurrent rendering, efficient Virtual DOM reconciliation.
Language	TypeScript 5.7.2	Strict static type safety, compile-time error elimination, explicit domain modeling.
Build Tool	Vite 6.4.3	Instant Hot Module Replacement (HMR), optimized Rollup tree-shaking, fast production bundling (<4s).
Styling & UI	Tailwind CSS 3.4.17 + PostCSS	Utility-first responsive styling, zero CSS bloat via PurgeCSS, mobile-first breakpoints.
Iconography	Lucide React 1.16.0	Crisp, tree-shakeable SVG icon set with consistent visual ergonomics.
QR Engine	qrcode 1.5.4	Client-side high-density dynamic QR code generation with error correction level M.
Celebration FX	canvas-confetti 1.9.4	Hardware-accelerated particle animation celebrating verified student submissions.
PWA Ecosystem	Service Worker + Web App Manifest	Offline app shell caching, standalone home-screen installation on iOS, Android, and Desktop.
PDF Engine	Native Print DOM Engine	Zero-dependency, vector-sharp PDF generation using A4 print styling and iframe sandbox.
Data Layer	Reactive LocalStorage Store + Firebase Bridge	Cross-tab event bus synchronization, automatic schema migrations, Firestore production rules.
Deployment	Vercel Production Edge	Global CDN distribution, automatic SPA route rewriting via vercel.json .

2.2 Architectural Flow

Architecture Philosophy: Decentralized Verification with Centralized Audit

The teacher creates the attendance session, which acts as the authoritative time-bounded anchor. The student client performs multi-factor biometric and spatial verification locally, submitting an immutable cryptographic proof that is atomically validated and permanently recorded in the institutional audit ledger.

CHAPTER 3

Multi-Layer Anti-Fraud Pipeline

3.1 Layer 1: Dynamic Short-Lived QR Tokens

To thwart screenshot-based proxy attendance, AttendIQ generates rotating time-bound tokens rather than static URLs. On the faculty display, the QR token rotates automatically every **25 seconds**.

- **Token Structure:** `ATTENDIQ:{sessionId}:{timestamp}:{salt}` encoded with cryptographic entropy.
- **Quick-Code Support:** Generates a companion 6-character alphanumeric quick-code for high-entropy manual entry in case of camera hardware limitations.
- **Grace-Period Tolerance:** The verification service implements a **30-second clock-drift window**, permitting students who scanned 2 seconds before rotation to validate cleanly while rejecting screenshots older than one cycle.

3.2 Layer 2: Haversine Campus Geofencing

The system queries device GPS coordinates via `navigator.geolocation.getCurrentPosition` using high-accuracy mode. Distance between the student's current position and the lecture hall is computed using the spherical **Haversine Formula**:

```
a = sin2(Δφ/2) + cos(φ1) · cos(φ2) · sin2(Δλ/2)
c = 2 · atan2( √a, √(1-a) )
d = R · c    (where R = 6,371,000 meters)
```

The default campus geofence is anchored at **26.9855° N, 75.7725° E** (SBCET Campus, Benad Road, Jaipur) with an institutional radius of **150 meters**.

Location Permission Denial Recovery Protocol

If a student denies browser location access or is inside a shielded basement lab where GPS signals attenuate, AttendIQ activates the **Campus Wi-Fi Beacon Fallback**. The student connects to the authorized institutional SSID (`SBCET-STUDENT-5G`) and validates through the college subnet gateway.

3.3 Layer 3: Biometric Face Match & Liveness Detection

To guarantee that the student holding the device matches the enrolled roll number, AttendIQ accesses the front-facing webcam stream inside a facial framing viewport:

- **Facial Verification:** Live canvas frame capture compares student facial landmarks against enrollment biometric profiles stored during admission.
- **Interactive Anti-Spoofing Liveness Challenge:** The user is prompted with randomized interactive challenges: *"Please blink both eyes naturally now"*, *"Smile at the camera"*, or *"Turn your head slightly to the left"*. Temporal frame analysis verifies motion, defeating static photo or pre-recorded video presentation attacks.

3.4 Layer 4: Atomic Duplicate Prevention Lock

To eliminate double submissions, the database enforces an atomic composite key: `rec_{sessionId}_{studentId}` . Once written, subsequent attempts within the same session throw a `DUPLICATE_ATTENDANCE_LOCK` exception.

CHAPTER 4

Progressive Web App (PWA) Implementation

AttendIQ is fully configured as an installable Progressive Web App (PWA), providing students and faculty with a native app experience on iOS, Android, macOS, and Windows without requiring app store distribution.

4.1 Web App Manifest (`public/manifest.json`)

```
{
  "name": "AttendIQ – Smart College Attendance System",
  "short_name": "AttendIQ",
  "start_url": "/",
  "display": "standalone",
  "orientation": "portrait-primary",
  "background_color": "#020617",
  "theme_color": "#4f46e5",
  "icons": [
    { "src": "/icons/icon-192.png", "sizes": "192x192", "type": "image/png" },
    { "src": "/icons/icon-512.png", "sizes": "512x512", "type": "image/png", "purpose": "any maskable" }
  ]
}
```

4.2 Service Worker Caching Architecture (`public/sw.js`)

The service worker employs a dual-strategy caching pattern:

- **Navigation Requests (HTML shell):** `Network-First` strategy falling back to the precached `/index.html` shell when college Wi-Fi experiences network dropouts.
- **Static Bundles (JS, CSS, Icons, Fonts):** `Stale-While-Revalidate` strategy ensuring instant UI load times while background-updating assets on subsequent visits.

4.3 Apple Mobile & Home-Screen Integration

Configured in `index.html` with `apple-mobile-web-app-capable`, `apple-mobile-web-app-status-bar-style="black-translucent"`, and iOS touch icons, enabling seamless 1-tap installation via Safari's "Add to Home Screen".

CHAPTER 5

Database Schemas & Institutional Seed Data

5.1 Core Domain Entities

ENTITY	KEY ATTRIBUTES	INSTITUTIONAL ROLE
Department	id, name, code	First Year Engg, CSE, CS, AI, CSE-DS, Mechanical Engg.
Section	id, name, semesterId, courseId	Section A (CSE & CS, 65 students), Section B (AI, DS, ME, 49 students).
Subject	id, name, code, credits, status	CY101 (Chemistry), MA101 (Math), CS102 (PPS), EC101 (DE), HS101 (Comm).
TimetableSlot	dayOfWeek, startTime, endTime, teacherId	Weekly recurring slot mapped to lecture halls and professors.
AttendanceSession	id, teacherId, subjectId, currentQrToken	Time-bounded 10-minute active attendance window.
AttendanceRecord	studentId, sessionId, verification, status	Atomic submission record with GPS distance, face score, timestamp.
AuditLog	actorId, action, oldValue, newValue, reason	Immutable record of administrative changes and manual corrections.

5.2 Real SBCET Institutional Dataset

114

STUDENTS ENROLLED

11

ASSIGNED FACULTY

60

CONDUCTED SESSIONS

6

ACADEMIC BRANCHES

- Section A (65 Students):** Batch A1 (33 CSE students, Roll: 26SBCETCSE001 – 033) and Batch A2 (22 CSE + 10 CS students).
- Section B (49 Students):** Batch B1 (36 AI students, Roll: 26SBCETAI001 – 036) and Batch B2 (7 CSE-DS + 6 Mech students).
- Assigned Faculty:** Dr. Aastha Pareek (Chemistry), Dr. A. H. Khan & Dr. Vishal Sexena (Mathematics), Dr. Pankaj Meel (Communication Skills), Mr. Vikas Singh (PPS), Mr. Sikander Khan (Cyber Security), Mr. Happy Dabla & Dr. Syed Firoz Haider (Digital Electronics), Mr. Toofan Mukharjee, Mr. Vijay Sharma, and Dr. Nisha Poonia.

CHAPTER 6 Teacher Command Center & Day-Wise Timetable

The Faculty portal in AttendIQ is organized into three major workflows: Day-Wise Lecture Scheduling, Live Attendance Broadcasting, and Historical Course Registers.

6.1 Day-Wise Teaching Schedule Navigator

Rather than displaying an unorganized list of classes, `TeacherDashboard.tsx` features an intuitive day-selector tab interface (Monday through Saturday, plus an "All Days" view). When selected:

- Classes for that specific day are loaded in chronological order by start time.
- Displays Subject Code, Full Name, Assigned Section, Classroom, and Period Duration.
- Indicates real-time status: *Scheduled*, *Live Session Active*, or *Completed*.
- One-tap button to **"Start Attendance"** or **"Resume Live Screen"**.

6.2 Live Attendance Session (`TeacherLiveSession.tsx`)

- **Dynamic QR Display:** Center-stage high-contrast QR code rotating every 25 seconds with an animated countdown ring.
- **Live Attendance Counter:** Real-time display showing present count vs. total section enrollment (e.g. `62 / 65 Present – 95.4%`).
- **Real-Time Stream Feed:** Instant animated ticker showing students checking in with verification metadata (QR Valid, GPS Distance, Face Match Score).
- **Instant PDF Export Button:** Direct header action to print or save the session roll call sheet as PDF right from class.

CHAPTER 7

Automated PDF Reporting Engine

In response to faculty requirements for archival and statutory RTU compliance, AttendIQ includes a client-side **PDF Generation Service** (`src/services/pdfService.ts`) and modal preview component (`src/components/PdfPreviewModal.tsx`).

7.1 Single Session Attendance Roll Call Sheet

- **Institutional Letterhead:** Sri Balaji College of Engineering & Technology, Jaipur with AICTE and RTU Kota credentials.
- **Lecture Metadata:** Subject Name & Code, Section, Faculty Name, Date, Time Slot, Lecture Hall, and Total Enrolled vs. Present counts.
- **Full Student List:** Every student in the section is itemized with Roll Number, Name, Branch, Batch, Status (bold green **PRESENT** or bold red **ABSENT**), check-in timestamp, and verification details.
- **Institutional Signatures:** Three-way sign-off blocks for **Subject Faculty**, **Class Coordinator**, and **Head of Department (HOD)**.

7.2 Cumulative Subject Register & Shortage Report

- Generates complete semester-long course registers with total lectures conducted, total lectures attended per student, attendance percentage, and prominent **SHORTAGE (<75%)** warning flags.

CHAPTER 8

Student Experience & Scan Wizard

The student interface is mobile-first, lightweight, and structured to guide the student through the four verification checks in under 10 seconds:

8.1 Multi-Step Verification Flow (`StudentScanWizard.tsx`)

1. **Step 1: Session Discovery & QR Scan:** Automatically identifies active faculty broadcasts or allows camera scanning of the dynamic QR.
2. **Step 2: Campus Geofence Validation:** Reads GPS coordinates and computes distance to the classroom anchor. Displays distance in meters (e.g. *"18m from SL1 Lecture Hall — Verified"*).
3. **Step 3: Biometric Face & Liveness Match:** Activates front camera with animated framing oval. Analyzes facial features and challenges the student to blink or smile.
4. **Step 4: Digital Attendance Receipt:** Triggers confetti celebration with digital verification receipt, cryptographic transaction hash, and updated subject attendance percentage.

8.2 Analytics & Shortage Warning Gauge

The student dashboard features a circular SVG attendance gauge. If the student's overall or subject-wise attendance drops below 75%, an automated amber/red shortage alert is rendered, detailing exactly how many consecutive classes must be attended to regain university eligibility.

CHAPTER 9

Administrator Console & Audit Trail

The Administrator console provides the Registrar and Head of Department with full governance over college structures:

- **Academic Hierarchy Management:** CRUD controls for Departments, Courses, Semesters, Sections, Subjects, and Classrooms.
- **User Directory:** Searchable directory of 114 students and 11 faculty members with branch, batch, and biometric status filters.
- **Campus Geofence Configuration:** Visual map coordinates, latitude/longitude adjustment, and radius slider (50m to 500m).
- **Immutable Audit Log:** Searchable audit trail recording every session opening/closure, rule change, and attendance correction with mandatory justification reasons.
- **CSV Data Export:** One-click sanitization-hardened CSV export for university registrar filings.

CHAPTER 10

Security Hardening & Future Roadmap

10.1 Security Vulnerability Mitigations

VULNERABILITY VECTOR	POTENTIAL IMPACT	ATTENDIQ ENGINEERED DEFENSE
QR Screenshot Proxy	Students forwarding photos to absent peers.	25-second rotating cryptographic tokens + 30s grace window.
Off-Campus Submissions	Students marking attendance from hostels.	Haversine GPS distance calculation + Campus Wi-Fi Beacon check.
Photo Presentation Attack	Holding a printed photo to the camera.	Interactive random liveness challenges (blink, smile, turn).
CSV Formula Injection	Arbitrary code execution in Excel (CWE-1236).	All exported cells starting with <code>=</code> , <code>+</code> , <code>-</code> , <code>@</code> escaped with single quotes.
Double Submissions	Network race conditions marking multiple times.	Atomic composite key lock (<code>{sessionId}_{studentId}</code>).

10.2 Production Build Verification

```
> attendiq@1.0.0 build
> tsc -b && vite build

vite v6.4.3 building for production...
✓ 1969 modules transformed.
rendering chunks...
dist/index.html          1.66 kB | gzip:  0.81 kB
dist/assets/index-CUPSancM.css 47.58 kB | gzip:  8.36 kB
dist/assets/index-BpMz1i-s.js 508.76 kB | gzip: 124.22 kB
✓ built in 9.36s with 0 errors
```

10.3 Future Engineering Roadmap

- **Hardware Bluetooth Low Energy (BLE) Beacons:** Installing BLE beacons at lecture hall doors for zero-touch physical presence validation.
- **Automated Guardian SMS/WhatsApp Alerts:** Integrating Twilio / Gupshup gateways to automatically notify parents when attendance drops below 75%.
- **Predictive Shortage Analytics:** Machine learning models forecasting semester shortfall risk based on early attendance trends.

INSTITUTIONAL SIGN-OFF & PROJECT ACCEPTANCE

Project Lead / Developer

AttendIQ Development Team

Head of Department (First Year)

Dr. Aastha Pareek / Dr. A. H. Khan

Principal & Registrar

Dr. Surendra Singh

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